Siddaganga Institute of Technology

(An autonomous institution affiliated to VTU, Belagavi, Approved by AICTE, New Delhi, Accredited by NAAC with 'A++' grade & ISO 9001:2015 Certified)

Department of Computer Science & Engineering

(Program Accredited by NBA)

ARTIFICIAL INTELLIGENCE & DATA SCIENCE

AND

ARTIFICIAL INTELLIGENCE & MACHINE LEARNING

DATA ANALYTICS WITH EXCEL AND R LABORATORY (S4CCSA05)

Laboratory Cycles and Evaluation Scheme - 2025-26 (Even Sem)

Lab Hours/ Week	: 2	Credits:	1
CIE Marks:	50	SEE Marks:	50

Guidelines to the students:

- Students are assessed every week for 35 marks.
- The students must complete implementation of the respective cycle programs on or before mentioned deadline.
- The students must carry an observation book to the lab which contains programs to be executed along with algorithms required for current lab set program.
- Students must maintain a file which includes the evaluation sheets and printouts of successfully executed programs with proper inputs and outputs and related information about each program as per the format provided.
- CIE marks for the practical course is computed by adding the average of the marks secured by the student for conducting each of the experiments with the marks secured in the test conducted and also the marks secured for the open-ended experiments at the end of the course. Two announced tests will be conducted. The syllabus and marks for each test is given below:

Activity	Syllabus	Marks
TEST	DAE + DAR	(T1(15) + T2(15))/2
Open Ended Experiments	Excel with VBA and R/Python/SQL integration	10

Lab test schedule:

Test-1: April 2nd Week (offline Mode)
Test-2: May 4th Week (offline Mode)
Demo of open-ended experiments: June 1st Week (offline Mode)

Evaluation Scheme:

Break-up of CIE marks for practical courses with Open Ended Experiments

Details	***Marks***
Regular Lab Work and writing lab records	(15+10) 25 marks
Lab test and Viva-voce at the end of the semester	(10+5) 15 marks
Evaluation of open ended experiment	10 marks
CIE	**50 marks**
SEE	**50 marks**
Total	**100 marks**

- **Evaluation scheme for Continuous Internal Evaluation (CIE): 50 Marks**

Break up of CIE marks:

Average marks of Weekly evaluation	25 Marks
Lab test Average	15 Marks
Demo Open Ended Experiments	10 Marks
Total	**50 Marks**

- CIE is computed by adding the average of the marks secured in weekly evaluation plus the average of marks secured in tests.
  - Evaluation scheme for Test :(15 Marks)
    - Write Up – 5 Marks
    - Execution – 5 Marks
    - Results with modification – 5 marks
- **Evaluation scheme for weekly evaluations:(25 Marks)**

Every week the students are assessed for 35 Marks based on the following Rubrics:

File maintenance = 08 marks				
Maintaining file (08 Marks)				
Excellent	Very Good	Good	Average	Poor
8 Marks	*7 marks*	*5-6 Marks*	*2-4 marks*	*1 mark*

Implementation and execution of the programs = 12 marks									
Implementation of complete program (7 Marks)					**Execution of programs for all possible inputs (5 Marks)**				
Excellent	Very Good	Good	Average	Poor	Excellent	Very Good	Good	Average	Poor
7 marks	*6 marks*	*4-5 marks*	*2-3 marks*	*1 mark*	*5 marks*	*4 marks*	*3 marks*	*2 marks*	*1 mark*

Answering 100% of Questions asked			
Excellent	Good	Average	Poor
5 marks	*4 marks*	*2-3 marks*	*1 mark*

  - Evaluation scheme for Open Ended evaluations:(10 Marks)

Excellent	Good	Average	Poor
9-10 marks	*7-8 marks*	*5-6 marks*	*1-4 marks*

Evaluation Scheme for SEE:

- In SEE all question Carries equal marks.
- Students are required to execute one program from lab set problems.
- The questions are allotted based on lots.
- SEE is evaluated for 50 marks using the following rubrics.
- Semester End Evaluation (SEE): 50 marks.
    - Write Up – 10 Marks
    - Execution – 20 Marks
    - Results with modification – 10 Marks
    - Viva – 10 Marks
- On taking change of program, questions from lab set will be changed and the questions are evaluated for **80%** of marks only.

Course outcomes (COs):

CO 1	**Build** Spreadsheet application by creating structured dataset
CO 2	**Automate** the problem solutions using Excel Formulas, Excel advanced features and VBA Macros
CO3	**Apply** analytics using R programming concepts
CO 4	**Analyze** the large and complex data set using Excel with integrated tools such as R, Python, SQL, SPSS etc.

Mapping of Course outcomes (COs) to Program outcomes (POs):

Program Articulation Matrix															
Course Outcomes	**Program Outcomes**												**Program Specific Outcome**		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1			2												2
CO2			2												2
CO3			2												2
CO4				2	2			2							2
over all			2	2	2			2							2

Degree of compliance 1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

Experiments:

Deadline: 01-06-2026

		Mapping		
		CO	**PO**	**PSO**
1.	Create a spreadsheet for 100 students records which consist of following attributes Student USN, Student Name, course taken , Marks Obtained, Grade points	1,2	3	3

a. **Create a suitable examination database and find the sum of the marks(total) of each Student and Average**

b. Obtain **Letter grades, grade points and corresponding marks range as per below table:**

Level	Out-standing	Excellent	Very Good	Good	Above Average	Average	Pass	Fail
Letter Grades	O	A+	A	B+	B	C	P	F
Grade Points	10	9	8	7	6	5	4	0
Absolute Marks Range (%)	≥90	<90 - ≥80	<80 - ≥70	<70 - ≥60	<60 - ≥55	<55 - ≥50	<50 - ≥40	<40
	(90-100)	(80-89)	(70-79)	(60-69)	(55-59)	(50-54)	(40-49)	(0-39)

NE: If a student fails to satisfy attendance (85%) and/or CIE (40%) requirements for the course/s then such course/s shall be marked as NE i.e., Not Eligible to appear for SEE in that course/s.

F' Grade is awarded under the following conditions:

 Failing to secure 35% of marks in SEE (min. 35/100)

c. **Calculate CGPA and SGPA** as per below rule

The academic performance of a student is indicated by two different indices, Semester Grade Point Average (SGPA) and Cumulative Grade Point Average (CGPA).

- SGPA is an indication of the performance of the student in the current semester.

  SGPA is calculated as indicated below.

$$\textbf{SGPA} = \frac{\Sigma\ [(\text{Course credits}) \times (\text{grade points})] \text{ for all courses with letter grades (in that semester)}}{\Sigma\ [(\text{Course credits})] \text{ for all courses registered in that semester including F grades and excluding W and DP courses.}}$$

- CGPA is an indication of the cumulative performance of the student from the first semester up to the current semester.

  CGPA is calculated as indicated below.

$$\textbf{CGPA} = \frac{\Sigma\ [(\text{Course credits}) \times (\text{grade points})] \text{ for all the courses whose letter grades are E and above from the I semester till the current semester.}}{\Sigma\ [(\text{Course credits})] \text{ whose letter grades are E and above from the I semester till the current semester.}}$$

Calculate SGPA and CGPA for 1st semester, IInd Semester, Supplementary Semester, and for the entire course.

d. **Calculate Percentage equivalence of CGPA. Apply the formula Percentage of Marks = (CGPA - 0.75)x10**

e. **Calculate CGPA range and corresponding class as per below table**

CGPA	Class
≥5.00 & < 6.75	Second
≥6.75 & < 7.75	First
≥7.75	Distinction

Visualization:

a. Freeze rows and columns
b. Sort the data according to USN.
c. Remove duplicates.
d. Filter: Distinction, First and Second-class data
e. Format the data using different table styles, conditional formatting including colours, labels, or data bars etc.
f. Visualize the data using Charts and Perform result analysis using Grade against Each Subject. Find cases where majority of total value attributed to failure, second class, First class and distinction.
g. Assuming that F grade students are passed in the exam in subsequent semester attempt and modify the data accordingly. Visualize the same over time series of data and do outlier.
h. Apply the following formula suitably and analyse the outcome.
   - IS FUNCTIONS
   - CONDITIONAL FUNCTIONS
   - MATHEMATICAL FUNCTIONS

Additional – Identify and Learn the short cut key for using above Excel features

2. Create Sales Database (big data – min 100 records) to predict the sales of product against quantity sold, Net Value, Gross Value etc. Sample is shown below (100 rows, 32 columns). 1,2 3 3

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	P	R
1	TXN Type	TXN No	TXN Date	Year	Month	Month Name	Day	Day Name	Qtr	FY Qtr	Location Name	Salesman	Salesman Name	Customer	Customer Short Name	Cust Type Name
2	INV	173914	02-01-2023	2023	1	January	2	Wednesday	Qtr 1	FY Qtr 3	Warehouse	TNS92	Ram	HA2161	Raj	Modern Trade

Y	Z	AA	AB	AC	AD	AE
Item Name	Item Short Name	Item Brand Name	Item Type Name	Item Make	Item Make Name	Item Identific
MQ 100 - Electro Appliances HAND BLENDER - 22111020	MQ 100 - Electro Appliances HAND BLENDER -	Electro Appliances	Handheld Appliances	GER	Germany	Electro Appli

AK	AL	AM
Discount	Gross Value	Net Value
0	13600	13600

i. Apply the following formula suitably and analyse the outcome.
   - HLOOKUP, VLOOKUP
   - FIND, SEARCH
   - LEFT, RIGHT, MID, TRIM, CONCANTENATION
   - COUNT, COUNTA, COUNTBLANK
   - DATE and TIME

ii. Apply Conditional Formatting including labels, colors, data bars etc.

a. Calculate How much Sales for Item # (# can be any specific item in datasheet)
b. Calculate Total Sales in any specific location
c. Calculate Total Sales during week and weekends
d. Apply conditional formatting for quarter wise sales data per salesperson. Note: Sample is shown below

Salesman Name	Qtr 1	Qtr 2	Qtr 3	Qtr 4
Ram	2,37,389	2,97,355	1,91,538	1,46,848
Jyothi	1,96,959	4,33,803	1,52,971	3,64,452
Sree Rav	85,892	1,83,200	3,51,871	2,48,707
Jaya	58,688	1,11,398	17,209	1,01,830

e. Apply conditional formatting for analysing change in sale. Note: Sample is shown below

Salesman Name	Qtr 1	Qtr 2	1st Half Sales	Qtr 3	Qtr 4	2nd Half Sales	Change
Ram	2,37,389	2,97,355	5,34,744	1,91,538	1,46,848	3,38,386	-37%
Jyothi	1,96,959	4,33,803	6,30,762	1,52,971	3,64,452	5,17,422	-18%
Sree Rav	85,892	1,83,200	2,69,092	3,51,871	2,48,707	6,00,578	123%
Jaya	58,688	1,11,398	1,70,087	17,209	1,01,830	1,19,038	-30%

f. Represent the Sales gain (e.g., Red bar for -ve, Blue bar for +ve) through data bar. Note: Sample is given below

Invoice Date	Sales	Bar
12-02-2019	(12,456)	(12,456)
08-04-2019	47,335	47,335
25-09-2019	84,133	84,133

iii. Analyze the data by visualization (apply charts) for Average Price per Month. (Note: Obtain Average price by computing the ratio of net value over quantity)

Analytics and Visualization :

a) Format existing data table into pivot table

b) Summarize City and Sales. Sample is shown below.

a.

Cust City Name	Sum of Quantity	Sum of Net Value
Bangalore	15,348	23,27,133

c) Calculate the City and Sales Contribution. Sample is shown below.

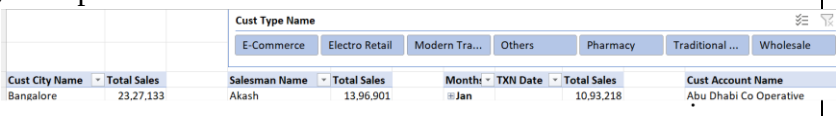
a.

Cust City Name	Total Qty	Total Sales	WOB
Bangalore	15,348	23,27,133	17.29%

d) Apply the slicers

e) Create multiple pivot table with reference to Customer Type, Salesman, Month wise sales transactions and Customer Account Name(Insert extra column by name Customer Account Name to the dataset). Sample is shown below.

a.



f) Outline form - Filter, Sort, Group and Apply advanced conditional formatting to the pivot table

g) Create Pivot Report - Analyse and Visualize sales

h) Create Pivot table with external data – show in tabular form

Use Slicers, auto filters etc

3.	Create worksheet with following fields: Empno, Ename, Basic Pay(BP), Travelling Allowance(TA), Dearness Allowance(DA), House Rent Allowance(HRA), Income Tax(IT), Provident Fund(PF), Net Pay(NP). Apply appropriate Excel formulas to, a) Calculate DA , where DA = 50% of basic Analyze and Visualize, total DA obtained in FY. Apply the conditional formatting for highest DA paid for an employee. b) Calculate Total HRA received in a given FY Analyze, How much HRA can be claimed? The HRA exemption is calculated as the minimum of the following three amounts: i) Actual HRA received from the employer ii) Rent Paid – 10% of Basic Salary Visualize HRA claims of employee. Apply conditional formatting for	1,2	3	3

minimum and maximum claims.

c) Calculate Income Tax

The tax rate for a salaried individual (under 60 years) as per the old regime is as follows:-

The tax rates for FY 2023-24, as per the new tax regime is, as follows:-

Tax Slab	Tax Rate
Upto Rs 3,00,000	Nil
Rs. 3,00,001-6,00,000	5%
Rs. 6,00,001-9,00,000	10%
Rs. 9,00,001-12,00,000	15%
Rs. 12,00,01-15,00,000	20%
Above Rs. 15,00,000	30%

Analyze how to save Income tax for a salaried individual?

To reduce taxes, it is highly important to understand the salary structure:-

- Taxable Salary Income = Salary (-) Exemptions (Exempt allowances + Standard deduction +Profes
- Net taxable income= Taxable Salary Income (-) Deductions under chapter VIA

Visualize the deductions and apply the conditional formatting appropriately

d) Calculate Provident Fund

Assume, Employees' portion of the EPF = 12% * (Basic Salary + DA)

Analyze and Visualize,

i) Employee contribution and Employer Contribution for Basic + DA <= Rs. 12000

ii) Employee contribution and Employer Contribution for Basic + DA >= Rs. 12000

Sample is shown below:

Employee Provident Fund Calculator (EPF)

Employee Name:			
Rate of increase of Basic Pay:	5.00%	Interest Rate	8.65%
Your Contribution:	12.00%	CTC	30000
Company's Contribution:	3.67%	Basic Pay + DA:	12000

Year	Opening Balance	Basic Pay	Employee Yearly Contribution	Company Yearly Contribution	Total Contribution	Rate of Interest	Closing Balance
1	0.00	12000.00	17280.00	5283.76	22563.76	8.65%	24515.53
2	24515.53	12600.00	18144.00	5548.00	23692.00	8.65%	52377.48

e) Calculate Net pay.

Net Salary: Gross Salary – Deductions

Apply conditional formatting for highest net salary and lowest net salary and visualize the data.

4.	i) Animate the charts using VBA macro Consider Sales details for the year 2024 and 2025. Develop animated Bar chart with data range of Month on X – Axis and Sales on Y –Axis correspondingly. The chart should contain title, label, and legends. ii) Automated Gantt Chart and Cost Estimation Generator using VBA	2	3	3

	in Excel. The objective is to automate the creation of a Gantt chart and total project budget estimation from a task list using a VBA macro. The macro should dynamically create a timeline, highlight task durations in a Gantt-style format, and calculate the overall project cost. Analyze the timeline and cost obtained			
5.	Assess the Financial Statement of an Organization being supplied with 2 vectors of data: Monthly Revenue and Monthly Expenses for the Financial Year. You can create your own sample data vector for this experiment) Calculate the following financial metrics: a. Profit for each month. b. Profit after tax for each month (Tax Rate is 30%). c. Profit margin for each month equals to profit after tax divided by revenue. d. Good Months – where the profit after tax was greater than the mean for the year. e. Bad Months – where the profit after tax was less than the mean for the year. f. The best month – where the profit after tax was max for the year. g. The worst month – where the profit after tax was min for the year. Note: a. All Results need to be presented as vectors b. Results for Dollar values need to be calculated with $0.01 precision, but need to be presented in Units of $1000 (i.e 1k) with no decimal points c. Results for the profit margin ratio need to be presented in units of % with no decimal point. d. It is okay for tax to be negative for any given month (deferred tax asset) e. Generate CSV file for the data	3	3	3
6.	**Employee Productivity** Create a data frame with employee ID, name, department, age, and productivity score. a) Subset employees from the Sales department over 40 with a productivity score above 80. b) Merge with a second data frame containing employee ID and weekly working hours. c) Find correlation between age and productivity (Pearson/Spearman). Plot and log-transform the data.	3	3	3
7.	**Student Performance in Analytics** Create a data frame with student ID, name, course marks, grade, and project score. a) Filter students with project score > 85 and grade 'A'. b) Merge with attendance data using student ID. c) Identify top predictors of course marks and handle multi collinearity.	3	3	3
8.	**Online Course Platform** Create a data frame with course ID, title, enrollment, rating, and instructor. a) Filter courses with rating > 4.5 and enrollments > 1000. b) Merge with course ID and completion rate. c) Visualize relationships, remove correlated variables like rating and completion rate.	3	3	3
9.	**Smart City Traffic Monitoring** Create a data frame with junction ID, vehicle count, average speed, and accident rate. a) Subset records with high traffic and low speed. b) Merge with road type using junction ID.	3	3	3

	c) Find important features for accident prediction and handle collinearity.			
Open Ended Projects: Data Analysis and Automation using Excel + R/Python/SQL/SPSS/VBA Students are informed to download xlsx or csv files with large dataset (both structured and unstructured data) from Kaggle or GitHub or any online repositories. Import the files and analyze the data using Excel formulas with VBA macros or R or Python or SQl or SPSS. Use Advanced features of Slicers, Auto Filter, Conditional Formatting and AI tools.		4	4,5,8	3

Section	Name and signatures of the Lab Coordinators	
AI & ML	Dr. Sumalatha Aradhya	
AI & DS	Mrs. Anupama B S	Signature of the HOD

VISION:

To nurture young Computer Science Engineers with a unique learning experience blended with spiritual values that promotes holistic development by building a strong teaching and research environment that is capable of responding to the challenges of the 21st century.

MISSION:

1. To prepare students for productive careers in industry and academia, through comprehensive Computer Science & Engineering based educational programs which focusses on technological competence and ethical values with social commitment.
2. To create research ambience that fosters problem solving skills to cater to the needs of industry and society resulting in scholarly publications in reputed journals and conferences
3. To collaborate with industry, research organisations and higher schools of learning for applied research exposure and capability to handle consultancy projects along with inculcating innovation and entrepreneurship skills.

PROGRAM EDUCATIONAL OBJECTIVES (PEOs):

- Pursue successful careers in State/National/Multi-National companies as software developers by following sound professional and ethical practices in various cadres in key areas like networking, web design, cloud computing, big data processing, IoT, e-commerce, information security and so on.
- Work effectively in multi-disciplinary and multi-cultural teams and demonstrate good soft skills.
- Pursue higher education for a successful career in industry/academics/ research.
- Pursue life-long learning, by anticipating trends in computer science and engineering, to excel in industry/academia or own a startup for a successful career as entrepreneur.

Knowledge and Attitude Profile (WK)

WK1: A systematic, theory-based understanding of the natural sciences applicable to the discipline and awareness of relevant social sciences.
WK2: Conceptually-based mathematics, numerical analysis, data analysis, statistics and formal aspects of computer and information science to support detailed analysis and modelling applicable to the discipline.
WK3: A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline.

WK4: Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline; much is at the forefront of the discipline.
WK5: Knowledge, including efficient resource use, environmental impacts, whole-life cost, re-use of resources, net zero carbon, and similar concepts, that supports engineering design and operations in a practice area.
WK6: Knowledge of engineering practice (technology) in the practice areas in the engineering discipline.
WK7: Knowledge of the role of engineering in society and identified issues in engineering practice in the discipline, such as the professional responsibility of an engineer to public safety and sustainable development.
WK8: Engagement with selected knowledge in the current research literature of the discipline, awareness of the power of critical thinking and creative approaches to evaluate emerging issues.
WK9: Ethics, inclusive behavior and conduct. Knowledge of professional ethics, responsibilities, and norms of engineering practice. Awareness of the need for diversity by reason of ethnicity, gender, age, physical ability etc. with mutual understanding and respect, and of inclusive attitudes.

PROGRAMME OUTCOMES (POs):

1. **Engineering Knowledge:** Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.
2. **Problem Analysis:** Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4)
3. **Design/Development of Solutions:** Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5)
4. **Conduct Investigations of Complex Problems:** Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions. (WK8).
5. **Engineering Tool Usage:** Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6)
6. **The Engineer and The World:** Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7).
7. **Ethics**: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. (WK9)
8. **Individual and Collaborative Team work:** Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.
9. **Communication:** Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences.
10. **Project management and finance**: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.
11. **Life-long learning**: Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change. (WK8)

PROGRAM SPECIFIC OUTCOMES (PSOs):

1. **PSO1: Computer based systems development:** Ability to apply the basic knowledge of database systems, computing, operating system, digital circuits, microcontroller, computer organization and architecture in the design of computer based systems.
2. **PSO2: Software development:** Ability to specify, design and develop projects, application softwares and system softwares by using the knowledge of data structures, analysis and design of algorithm, programming languages, software engineering practices and open source tools.
3. **PSO3: Computer communications and Internet applications:** Ability to design and develop network protocols and internet applications by incorporating the knowledge of computer networks, communication protocol engineering, cryptography and network security, distributed and cloud computing, data mining, big data analytics, ad hoc networks, storage area networks and wireless sensor networks.